Journal of Marketing Analytics
https://doi.org/10.1057/s41270-025-00450-2

ORIGINAL ARTICLE

Predictability & explainability of survival analysis in churn prediction

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Received: 10 May 2025 / Accepted: 28 October 2025
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Abstract
This study addresses customer churn prediction in contractual utility services by applying survival analysis models, which provide time-to-event insights beyond traditional machine learning approaches. Churn is defined as subscription cancellation within a given period. Unlike classification models that only indicate whether churn will occur, survival models estimate hazard rates, capturing how churn risk evolves over time. Our objectives are twofold: (1) to compare two survival models—the Cox Proportional Hazards (CPH) model and the Aalen Additive (AA) model—in identifying key drivers of churn, and (2) to demonstrate their interpretability in predicting churn timing for more effective customer intervention strategies. Experiments with data from a gas utility company show that survival models can successfully predict customer churn across products and contract types. By estimating individual-level risk profiles, these models highlight customers most likely to leave, enabling segmentation based on churn likelihood and timing. This provides actionable insights for designing targeted retention efforts. Overall, the study demonstrates the added value of survival analysis in churn prediction: it not only forecasts whether customers are at risk but also when churn is likely, supporting timely, tailored strategies that reduce attrition and strengthen customer retention.

Keywords Survival analysis · Customer churn · Customer retention

Introduction

Customer churn represents a significant financial threat to companies across all industries. According to Harvard Business Review, acquiring a new customer can cost five to twenty-five times more than retaining an existing one (Gallo 2014). This disparity highlights the critical importance of proactive retention strategies; failing to implement them can result in millions of dollars in losses. To mitigate this risk, companies must seize every opportunity to reduce customer churn, which occurs when customers leave subscription to a product before a contract ends (early churn), switch to a different product at renewal (internal churn), or depart the company entirely (external churn).

The reasons behind customer churn can vary widely, presenting businesses with the challenge of identifying the key factors that significantly influence churn. Predictive models offer a powerful tool to pinpoint these factors, such as income, customer tenure, and product-type, etc. (Sing et al. 2024), providing valuable insight into areas that require attention when developing retention strategies. Traditional machine learning models, such as Logistic regression and Logit Boost (Hemlata, et al. 2020) have been widely used to predict whether a customer will churn and to identify significant predictors like tenure, contract duration, etc. These models have provided businesses with enough foresight to design basic retention strategies centered around the binary outcome of whether a customer will churn or not.

However traditional machine learning models only provide a fixed-time prediction and fail to capture the dynamic changes in the likelihood of customer churn over time. Understanding these continual changes is crucial for firms aiming to implement effective retention strategies. For instance, business can segment customers at high risk of churning within a specific time frame, such as next few

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Published online: 22 December 2025

months, and focus efforts on that group for targeted interventions (Ghavami et al. 2023).

This research study explores the application of survival models in predicting customer churn likelihood. Unlike traditional models, survival models incorporate the dimension of time into predictions while also maintaining the explainability of factors influencing churn. A significant advantage of survival models is their ability to estimate the probability of customer churn at specific points in time. This insight is invaluable for designing targeted, customer-level retention strategies.

The study uses time-to-event data, which captures not only whether the event (churn) occurs but also the time elapsed from a defined starting point until the event. A defining feature of such data is *censoring*—for some customers, the event has not occurred by the end of the study, so their exact churn time is unknown. For example, a customer who cancels after 100 days has a time-to-event of 100, while one still subscribed after 180 days is censored at 180. Survival analysis methods are designed to handle both event times and censored observations without bias.

This paper contributes to the field of customer retention analysis by conducting extensive experiments to demonstrate the explainability and applicability of the Aalen Additive model and the Cox Proportional Hazard model in customer survival analysis. Our work highlights the significance of incorporating the duration leading up to customer churn as a crucial predictive feature, offering valuable insights into customer behavior over time. By utilizing these survival analysis models, we provide crucial information and factors that influence customer churn. By leveraging on these factors in influencing customer retention, businesses can design more effective and personalized retention strategies. Our models can be used to predict customer churn and to also identify actionable factors that can improve customer engagement and loyalty on time.

Literature review

Research on customer churn has largely emphasized binary classification models (e.g., logistic regression, tree ensembles) that predict whether a customer will churn at a fixed horizon. While useful for ranking risk, these models suppress the time-to-event dimension that operators need to plan outreach and capacity over the customer life cycle. In addition, many recent advanced models function as black boxes, and interpretability remains a significant challenge (Peng et al. 2023; Manzoor et al. 2024). Post-hoc explanation tools such as SHAP partially restore interpretability for complex models, but they carry heavy computational cost and rely on feature-independence assumptions that are often violated in correlated customer relationship management (CRM) data (Lundberg and Lee 2017; Aas et al. 2021; Maan and Maan 2023).

Survival (time-to-event) analysis addresses the timing problem directly by modeling hazards over tenure while handling censoring (Schober and Vetter 2018). In applied CRM studies, the Cox proportional hazards (CPH) model is the default because it yields policy-friendly, interpretable hazard ratio summaries—clear effect sizes that support explanation, communication, and auditability—and it can be fit without specifying a baseline hazard (Cox 1972; Wang et al. 2019). Its initial applications were predominantly in medicine, particularly studies of patient survival times (Matsuo et al. 2019; Yang et al. 2021). In recent years, the use of survival models, including CPH, has expanded to fields such as marketing, with notable applications in customer retention (Tristan 2020) and churn prediction (Yan et al. 2022).

However, the CPH model has a key limitation: the proportional-hazards (PH) assumption—that covariate effects remain constant over time—is often assumed rather than tested. When pricing, tenure, or service incidents induce time-varying effects or diverse use-case scenarios, unexamined PH violations can bias effect sizes and mis-time interventions. Diagnostics and remedies (e.g., tests based on Schoenfeld residuals or time-by-covariate interactions) are well established but are inconsistently reported outside biostatistics (Grambsch and Therneau 1994).

A complementary lens comes from the Aalen additive (AA) hazards framework, which relaxes the PH assumption and allows covariate effects to vary over time, with cumulative-coefficient plots that show when a predictor matters—early, late, transiently, or persistently (Aalen 1989; Aalen et al. 2008). AA is widely used in medical event-history analysis precisely to reveal dynamics that a multiplicative, time-constant specification can mask (Lim and Zang 2009; Abadi et al. 2011; Ataee et al. 2020; Dizaji et al. 2020). More recently, applications have appeared in education (Chen et al. 2018) and human resource management (Basha et al. 2023). Yet it remains uncommon in churn research, despite the operational premium on timing. In settings where PH is doubtful, AA can complement CPH by surfacing effect trajectories while keeping explanations model-intrinsic and audit-ready.

Beyond semiparametric models such as CPH and AA, machine-learning survival methods—e.g., Random Survival Forests (Ishwaran et al. 2008) and neural Cox variants (Katzman et al. 2018; and later extensions by Kvamme, Borgan, and Scheel in 2019)—model time-to-event data and often improve discrimination. However, as noted earlier, they do so at the expense of transparency and auditability, which are critical in regulated utilities and consumer



markets. This accuracy-interpretability trade-off further motivates frameworks that prioritize model-intrinsic explanations (e.g., hazard ratios, effect-over-time plots) and minimize reliance on computationally heavy post hoc tools such as SHAP.

Taken together, the literature signals four actionable shortcomings: (i) fixed-horizon classifiers rank who will churn but not when, limiting the timing of outreach; (ii) many advanced models function as black boxes, and post hoc tools such as SHAP add computational burden and rely on questionable independence assumptions; (iii) in survival modeling, CPH is common but its proportional-hazards assumption is often untested, risking biased effects and mistimed interventions; and (iv) the Aalen additive framework, which reveals time-varying effects, remains underused in churn prediction and customer management.

Our study is designed around these gaps. We pair CPH with AA to balance "interpretability" (hazard ratios) and "dynamic fidelity" (time-varying effects). We retain CPH for clear, audit-ready hazard-ratio summaries, explicitly run PH diagnostics, and use AA to surface time-varying effect trajectories via cumulative-coefficient plots. This dual-model design keeps explanations intrinsic, minimizes reliance on heavy post hoc tooling, and preserves the advantages of time-to-event analysis with censoring. Finally, in a contractual utility context with product-specific cohorts, we map estimated hazards to actionable outreach windows, so survival outputs translate to who/when/how guidance for retention teams.

Model formulation

In this section, we present the formulations of the two models central to our study – the Cox proportional hazards (CPH) model and the Aalen additive (AA) model.

Cox proportional hazards model

The Cox proportional hazards (CPH) model (Cox 1972) represents the hazard function (the instantaneous failure rate) in multiplicative form. Specifically, the hazard at time t is the product of an unspecified baseline hazard function of time and an exponential function of the explanatory variables with unknown regression coefficients. A key assumption – the proportional-hazards (PH) assumption is that covariate effects on the hazard are constant over time; this underpins the interpretation of hazard ratios (HRs) and the model's applicability. The CPH model can therefore be written as follows:

$$\lambda_i\left(t|X\right)=\lambda_0\left(t\right)e^{\beta_1x_{i1}+\beta_2x_{i2}+\cdots+\beta_px_{ip}}$$

where i is the observation index; $\lambda_i\left(t|X\right)$ represents the Cox Proportional Hazard function at time t given the covariates $X=[x_1,x_2,\ldots,x_p]$; $\lambda_0\left(t\right)$ is the baseline hazard function for the case when all the covariates are set to zero. $\beta=[\beta_1,\beta_2,...,\beta_p]$ is the coefficients vector.

Utilizing the preceding definitions, the hazard ratio for two customers, i and j, at time t is defined as

$$HR_{i,j}\left(t\right)=\frac{\lambda_i\left(t\right)}{\lambda_j\left(t\right)}=\frac{\lambda_0\left(t\right)e^{\beta_1x_{i1}+\beta_2x_{i2}+\cdots+\beta_px_{ip}}}{\lambda_0\left(t\right)e^{\beta_1x_{j1}+\beta_2x_{j2}+\cdots+\beta_px_{jp}}}$$

$$=e^{\beta_1(x_{i1}-x_{j1})+\beta_2(x_{i2}-x_{j2})+\cdots+\beta_p(x_{ip}-x_{jp})}$$

where the hazard ratio in the CPH model quantifies the effect of a covariate (or multiple covariates) on the risk of an event.

To optimize the coefficients, the partial likelihood function for the model must first be defined. The partial likelihood for a single observation (customer *i)* who is at risk of churning at time t is expressed as

$$L_i\left(\beta\right)=\frac{e^{X_i^T\beta}}{\sum_{j\epsilon R(t)}e^{X_j^T\beta}},$$

where, $R\left(t\right)$ is the set of individuals who are part of the risk set at time t and $e^{X_i^T\beta}$ is the **relative** risk of the individual who experiences churn at t.

Using the above definitions, the partial likelihood of the CPH model is given as follows:

$$L\left(\beta\right)=\prod_{i=1}^{n}\left(\frac{e^{X_i^T\beta}}{\sum_{j\epsilon R(t_i)}e^{X_j^T\beta}}\right)^{\delta_i}$$

We specify the function as the partial likelihood because CPH does not require the specification of the baseline hazard function $\lambda_0\left(t\right)$

In $L\left(\beta\right)$, if individual i contributes to the likelihood, then the indicator function $\delta_i=1$; otherwise, it is set to 0.

By taking the logarithm of the partial likelihood, we obtain the partial log-likelihood, which can then be maximized using the Newton–Raphson method. This process yields the coefficient estimates for β, capturing the effect of each covariate on the hazard rate.

The Aalen additive model

The second model used in the study is the Aalen Additive model, first introduced by Aalen (1980). This non-parametric model allows the regression coefficients to vary over time. As noted by (Henderson and Milner 1991), the

covariates in this model can be interpreted similarly to a linear regression model, and any changes in the coefficients are easily identified at each distinct survival time point. The multi-covariate AA model incorporates the effects of each covariate in the model through time-varying coefficients.

Consider a study where customer churn is the target variable, and each customer is characterized by $\boldsymbol{p}$ covariates. The hazard function of the Aalen Additive model, as further described by Lefebvre and Giorgi (2021), is defined as follows:

$$\phi_i\left(t|X_i\right) = \beta_0\left(t\right) + \beta_1\left(t\right)x_{i1} + \cdots + \beta_p\left(t\right)x_{ip},$$

where $X_i = [x_{i1}, x_{i2}, \ldots, x_{ip}]$ consists of the predictors for individual i at time t. $\beta\left(t\right) = [\beta_1\left(t\right), \beta_2\left(t\right), \ldots, \beta_p\left(t\right)]$ is the vector of the time-varying coefficients for the covariates at time t. $\phi_i\left(t|X\right)$ is the Aalen Additive hazard function at time t for the ith observation.

The Unregularized Aalen Additive model estimates the coefficients by minimizing the sum of squared residuals over time, expressed as

$$\widehat{\beta}\left(t\right) = \underset{\beta(t)}{arg\min}\sum_{i=1}^{n}\xi\left(T \geq t\right)\left[Y_i\left(t\right) - \left(\beta_0\left(t\right) + \sum_{j=1}^{p}x_{ij}\beta_j\left(t\right)\right)\right]^2$$

where $\xi\left(T_i \geq t\right)$ is an indicator function such that

$$\xi\left(T_i \geq t\right) = \begin{cases}1, & \text{if the customer churns at or before time } t\\ 0, & \text{otherwise}\end{cases}$$

and $Y_i\left(t\right)$ is the observed outcome of individual i at time t.

Using the above definitions, the estimated coefficients at time t for the given covariates under a Ridge regression regularized setting is given by:

$$\hat{\beta}\left(t\right) = \arg\min_{\beta(t)}\sum_{i=1}^{n}I\left(T \geq t\right)\left[Y_i\left(t\right) - \left(\beta_0\left(t\right) + \sum_{j=1}^{p}x_{ij}\left(t\right)\beta_j\left(t\right)\right)\right]^2 + \tau\sum_{j=1}^{p}\left|\left|\beta_j\left(t\right)\right|\right|^2$$

In this experiment, the ridge regression (L2 regularization) is used to prevent model overfitting and to mitigate multicollinearity by introducing an L2 regularization penalty to the cost function, promoting parsimony in covariate selection. The strength of regularization is controlled by the parameter τ where larger values increase penalization, shrinking coefficients toward zero. L2 regularization is preferred over L1 because the primary goal of this experiment is coefficient shrinkage while retaining all variables. L2 handles multicollinearity better when dealing with time varying coefficient context.

The same solution to the penalized least-squares problem can also be obtained using the regularized ordinary least squares regression solution in matrix form. This approach yields the vector of coefficient estimates at time *t,* given by

$$\widehat{\beta}\left(t\right) = \left(X(t)^T X\left(t\right) + \tau I\right)^{-1}X(t)^T Y\left(t\right)$$

where, $X\left(t\right)$ is the covariate matrix at time t, $Y\left(t\right)$ is the customer churn vector at time t, I is the identity matrix ensuring the regularization term is applied, $\tau > 0$ is the regularization parameter that controls the degree of penalization and $\widehat{\beta}\left(t\right)$ are the coefficient estimates at time t.

Materials and methods

For this study, we used a dataset from a utility company consisting of 3,667 rows and 69 columns. Each observation represents a customer's subscription to a utility service product in a given month, along with associated usage data, temperature information, utility market price, and other relevant details for that month. Because customers may hold continuous subscriptions and may subscribe across multiple product plans, customer IDs often appear more than once in the dataset. In our study, churn is defined as one of three events: leaving before a contract ends (early churn), switching to a different product at renewal (internal churn), or terminating all services with the company. The target variable, Churn-Flag, is set to 1 if any of these events occur. For churned customers, the dataset also records the churn date.

We model internal and external churn together because both reflect a failure to retain the customer's engagement with the focal product. Whether a customer leaves the company or moves to another internal plan, the underlying behavior – a decision to abandon the current service – is driven by similar factors such as dissatisfaction or the pursuit of greater value. This approach follows Blattberg et al. (2008), who model customer transitions across states (e.g., *Premium Subscriber*→*Basic Subscriber*→*Churned*) as part of one unified attrition process, recognizing that internal downgrades and full exits both reduce customer lifetime value.

From a behavioral standpoint, Prospect Theory (Kahneman & Tversky 1979) suggests that customers evaluate potential gains and losses relative to their current plan. The same psychological process applies whether they switch to a

competitor or to another plan within the company. Treating all churns uniformly therefore aligns with both managerial objectives and behavioral theory, offering a comprehensive view of customer switching behavior.

This dataset exhibits an unusual class distribution with 75% churn versus 25% non-churn, which differs from typical business scenarios where non-churn is the majority class. This is not the result of re-sampling or labeling but reflects the business reality captured in the complete customer records provided by the sponsoring company, covering all customers in the focal area over a 7-year period (2016–2023). Because of this atypically high churn rate, the business manager is particularly motivated to identify the key drivers of customer attrition and to develop predictive tools that can help the company anticipate and mitigate future churn. By analyzing this extreme case, the study provides valuable insights into customer behavior at critical decision points and offers data-driven strategies to strengthen customer retention.

From the original 69 variables in the dataset, 13 variables (11 predictive features plus event and customer duration variables) were selected through a systematic process:

1. Domain expertise consultation – The utility company team removed variables lacking practical relevance.
2. Statistical significance testing – Variables with a p-value less than 0.05 (via correlation or chi-square tests) were retained.
3. Multicollinearity assessment – VIF analysis removed highly correlated predictors.

Below is a list of descriptions for some of the key factors. The full list of these features is provided in the Appendix.

- Churn flag: event variable or response
- Customer tenure: duration (in months, t)
- Heating Degree Days (HDD65): number of days in a month when the temperature was below 65°F
- Product Type: type of service a customer subscribes to (six-month or twelve-month)
- Daily Energy Usage: average daily utility consumption by a customer in any given month. This variable can also be used to derive other features, such as average monthly usage during a product subscription.
- Price Switch Count: number of times a customer switches between products
- Automatic Contract Renewal: binary variable indicating whether the contractor was activated for automatic renewal or not.

For the AA model, to better demonstrate the time-varying effects of predictive variables, each quantitative feature was divided into four ordinal categories containing roughly equal numbers of observations. The resulting labels – very low, low, medium, and high – correspond to quartiles defined as follows: *very low* includes values below the 25th percentile, *low* spans the 25th ~ 50th percentiles, *medium* the 50th ~ 75th percentiles, and *high* the top quartile (≥ 75th percentile). In all analyses, the *very low* category serves as the reference group (baseline) for comparison.

This transformation converted the time-varying continuous variables into ordinal covariates, allowing the AA model to estimate the effect of each level on customer churn over time. Both time-varying and fixed covariates were incorporated into the AA model to capture their dynamic influence on the hazard function over the observation period.

By relying on rank ordering rather than absolute values, quartile-based discretization minimizes sensitivity to outliers and mitigates skewness in the data. This approach provides well-defined ranges. For example, in each utility usage range (low, medium, or high daily utility consumption), we enable intuitive interpretation of gas utility across different utility consumption patterns. Additionally, balanced group sizes facilitate fair comparisons across segments, yielding robust and actionable insights for customer segmentation strategies.

The three categorized time-varying covariates used in the AA model were then preprocessed using dummy encoding, with the category serving as the reference group. Each four-level variable was thus represented by three dummy variables. For example, the three-level HDD65 variable was coded as HDD65_low, HDD65_medium, and HDD65_high.

We employed dummy encoding (one-hot encoding with a reference category) for the categorical covariates. This approach is well suited for survival analysis because it preserves interpretability, with coefficients representing effects relative to a clearly defined baseline (Allison 2010; Therneau and Grambsch 2000). Unlike label encoding, which can impose an artificial ordinal structure, dummy encoding allows each category to capture its own distinct effect without assuming monotonicity. It also avoids potential bias and instability associated with more complex encoding methods such as target encoding (Kuhn and Johnson 2013). These properties make dummy encoding particularly appropriate for hazard models, where model transparency and auditability are essential.

In addition, the quantitative variables in the data set were standardized using the min–max normalization technique to scale their values between 0 and 1 before training the model.

The dataset was divided into training and testing sets using an 80:20 train-test split, with the holdout 20% reserved for evaluating model performance. During the training phase, five-fold cross-validation was applied to the training data

to validate the models and fine-tune the hyper-parameters as necessary.

During the model training phase, the full training dataset was initially used to establish a baseline performance measure. Subsequently, the dataset was segmented into two subsets based on product type (6-month and 12-month subscriptions), as the factor significantly influenced churn prediction accuracy. The impact of this segmentation on model performance will be demonstrated in the Results section of this report.

For the AA model, Ridge Regression (L2 regularization) was utilized to reduce overfitting and address multicollinearity during training. As discussed earlier, by shrinking correlated coefficients toward zero, L2 regularization stabilizes parameter estimates and enhances the interpretability of individual factor effects. Hyperparameter tuning was conducted to identify the optimal L2 regularization parameter (τ). The grid search involved determining the optimal hyperparameter value (τ) for each customer segment. The search involved using 10 values of τ selected from the list: [0.0001, 0.0004, 0.002, 0.01, 0.05, 0.20, 1, 5, 20, 100] such that the range of the values is from 0 to 100. The list is generated by creating 10 values evenly spaced on a logarithmic scale. Selecting values randomly on a linear scale is highly inefficient and often ineffective, as it can easily miss critical orders of magnitude, potentially leading to underfitting or overfitting. As supported by Géron, A. (2019) using a logarithmic scale ensures that each order of magnitude is considered, facilitating an exhaustive and efficient search across a wide range of values.

Combined with fivefold cross-validation, the best-performing configuration for the experiment was identified using the concordance index score metric. The concordance statistics quantify the level of agreement between predicted risk scores and the observed time until an event, commonly used in survival analysis (Therneau and Watson 2017). A higher concordance index score indicates better model performance, meaning that customers predicted to be at high risk of churning were likely going to churn. The value of τ that achieved the highest average concordance score during cross-validation was selected as the optimal hyperparameter for training and testing the AA model.

Subsequently, cumulative hazard rate curves and individual hazard rate curves were generated for randomly selected 6-month contract customers to capture their risk patterns both before and after their contracts ended. Similarly, hazard rate curves were produced for 12-month contract customers, offering insights into their risk trajectories during and after the completion of their contracts. Additionally, concordance index scores were calculated for both 6-month and 12-month customers using the test data, enabling the evaluation of model performance across the periods before and after their respective contracts.

Figures 1 and 2 are flow chart illustrations of the AA and CPH models applied in this study.

Result

Cox proportional hazard (CPH) model

This section presents the results obtained from the CPH model, applied to different datasets categorized by contract duration (e.g., customers with a 6-month contract before or after six months, customers with a 12-month contract before or after twelve months, etc.) Table 1 below summarizes the performance of the CPH model across these datasets, evaluated using concordance index scores.

According to Table 1, the CPH model demonstrates the strongest performance for customers with 12-month contracts during the active contract period, achieving a concordance index score of 79.1%**.** Similarly, customers with 6-month contracts before the contract end also show strong predictability (78.5%). However, performance declines noticeably once the contract periods expire, dropping to 71.8% for 12-month customers and 60.1% for 6-month customers after their respective contract terms. This decline indicates that customer churn behavior becomes significantly more unpredictable once contractual obligations end. The overall concordance index score for the full dataset is 73.3%, highlighting the model's reasonable predictive capability across diverse customer groups. Each case will be further examined in the following subsections.

Full Dataset (All Customers Included)

The dataset evaluates the model's performance across all customers subscribed to any product type, covering periods both before and after contract termination. A key question, explored later in this section, is how the model's performance differs when the dataset is segmented by product type.

Figure 3 below illustrates the survival rates for different product types using the full dataset. The survival rate declines more sharply during the first 12 months for the 6-month product type compared to the 12-month product type. This suggests that customers with 6-month contracts are more likely to churn when the contract ends (6th month) or immediately after (7th month), while customers with 12-month contracts experience a slower, more gradual decline. Notably, both product types exhibit significant drops in survival rates around the 6-month and 12-month marks after the contract begins, regardless of the contract length. This reflects actual customer behavior rather than a modeling artifact and underscores critical periods where targeted customer retention efforts could be most effective.

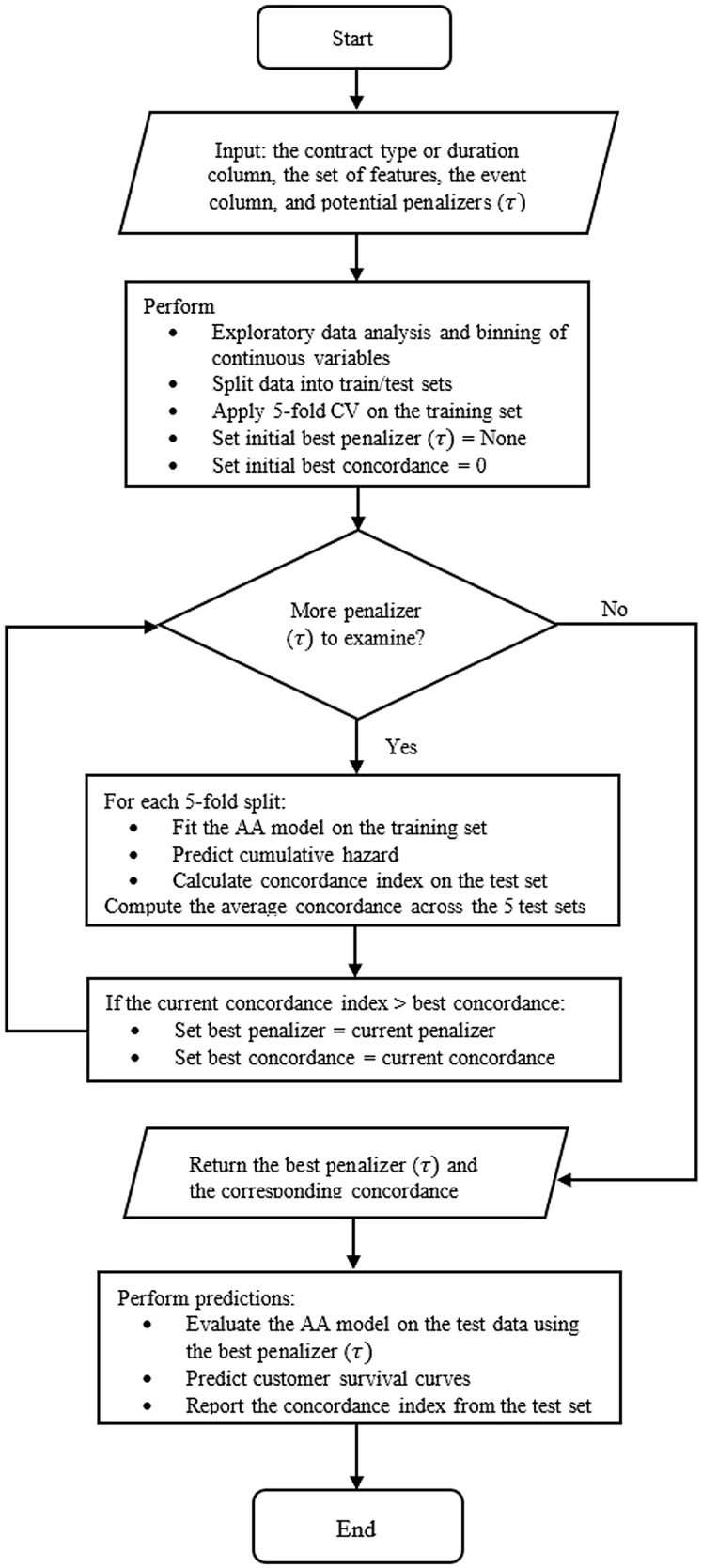


Fig. 1 The AA model flowchart

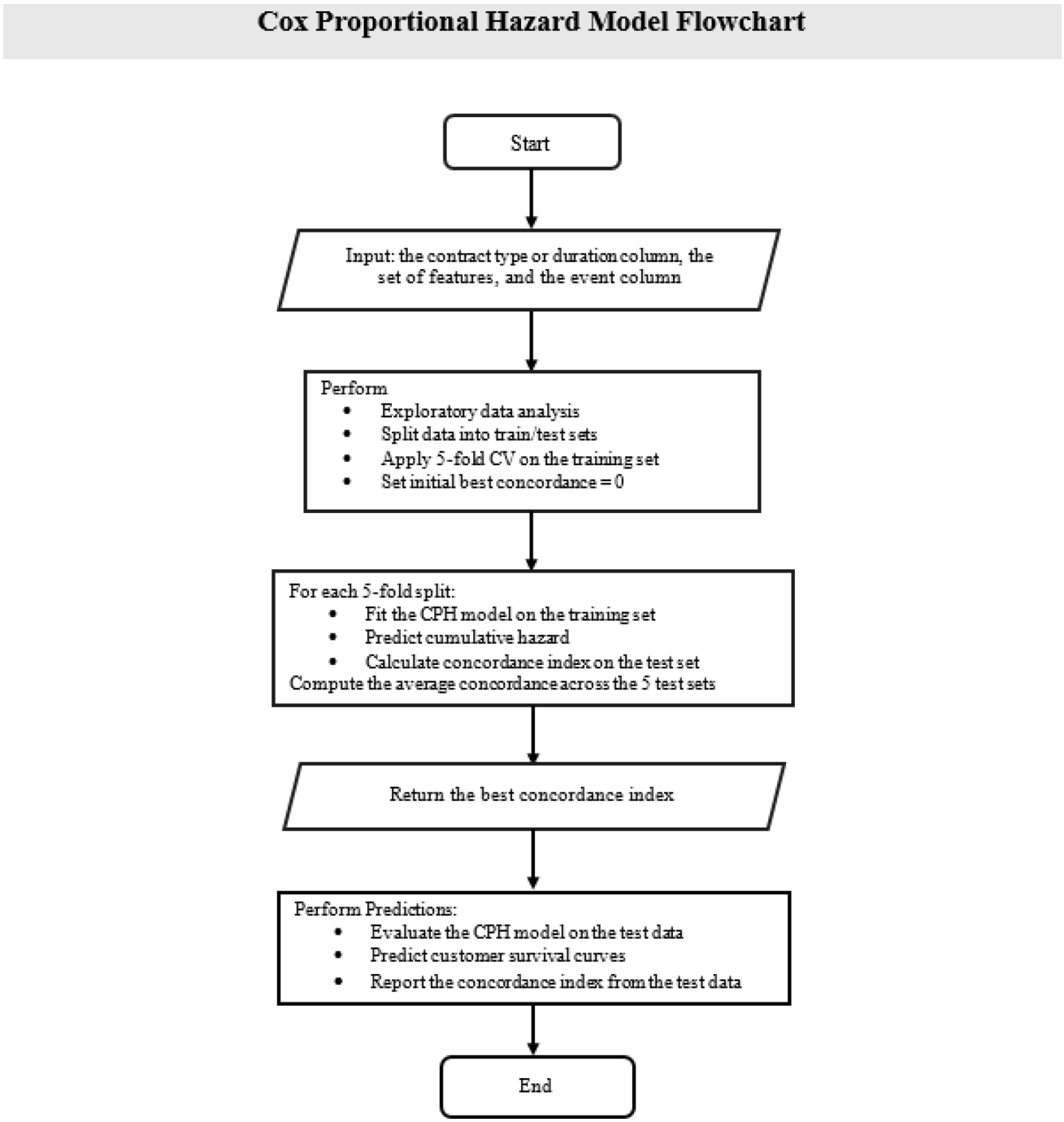


Fig. 2 CPH model flowchart

Table 1 CPH model performance summary table

Customer Groups	CPH Test Dataset Concordance Index Score
All customers (full dataset)	73.3%
6-month contract customers before the 6th month	78.5%
6-month contract customers after the 6th month	60.1%
12-month contract customers before the 12th month	79.1%
12-month contract customers after the 12th month	71.8%

Figure 4 below illustrates how churn survival rates are affected by equally binned average utility usage by product type. In the early months, customers with low utility usage tend to churn a lot faster compared to those with high usage. This observation motivates further analysis of the impact of average utility usage, a time-varying factor, on churn rates. It is important to note that the CPH model is not equipped to analyze such time-varying effects.



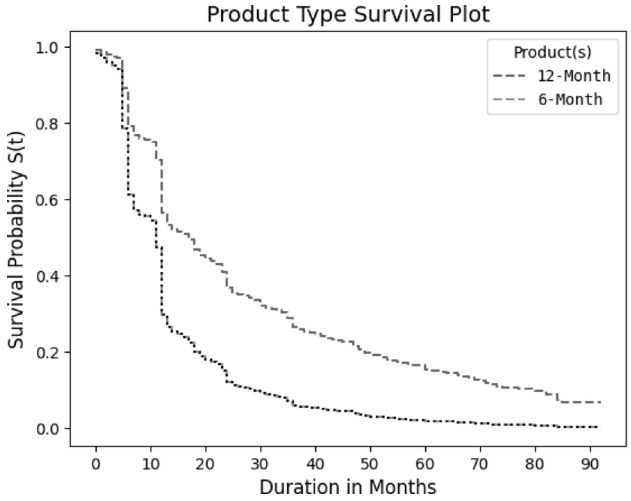


Fig. 3 Survival probability plot by product type

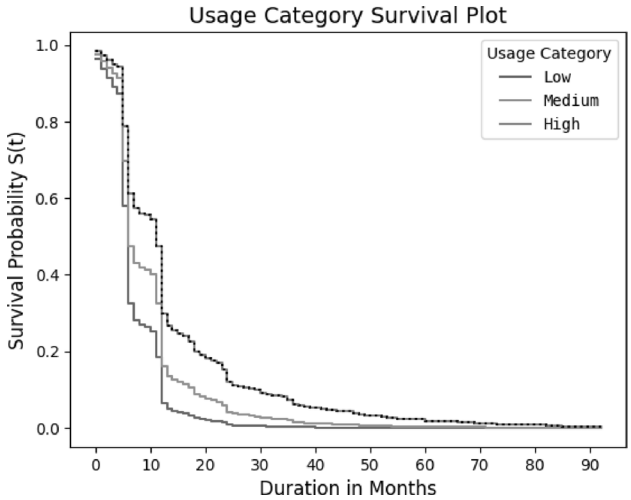


Fig. 4 Survival probability plot by usage category

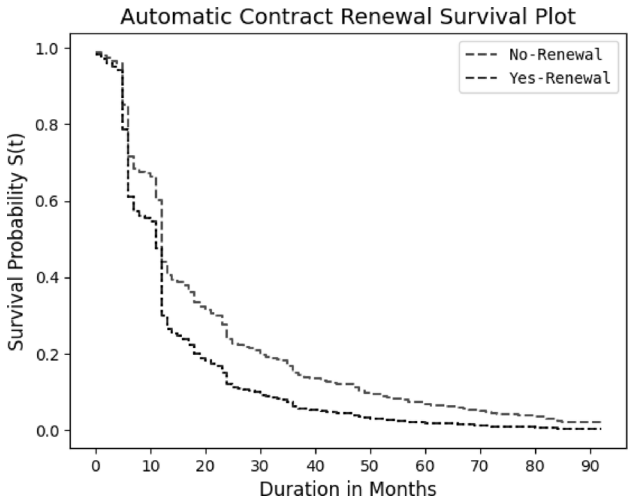


Fig. 5 Survival probability plot by contract renewal category

Figure 5 illustrates customer survival rates based on the contract renewal indicator. This factor records whether a customer had signed up the automatic contract renewal (ACR). Interestingly, customers with the ACR option activated exhibited lower survival rates on average. This trend may reflect a preference among these customers for maintaining control over their finances, avoiding the perception of an automated system making decisions, which can sometimes feel like a financial burden. In the long term, customers often either switch out of ACR or churn from the product entirely. This helps explain the sharp drop of survival rates around the six-month and twelve-month mark, corresponding to typical contract renewal or cancellation periods.

6-month product type

The Full Dataset model provided a comprehensive overview of model performance by using data from all product types. However, to gain deeper insight into customer behavior across different contract types, the dataset was segmented by the product type variable, which identifies the specific contract associated with each observation. For more detailed analysis, the 6-month dataset was further divided into two time periods: before the 6-month contract period and after it. The "after 6-month" group specifically includes customers who signed up for a 6-month contract but did not churn at the end of the contract period.

As shown in Table 1, compared to the Full Dataset model, performance improved significantly to 78.5% for the "before 6-month" subgroup but declined for the "after 6-month" subgroup. This inverted performance trend occurs because the 6-month product type model effectively captures actionable insights into customer behavior during the contract period, allowing for more accurate predictions before the 6-month contract ends. However, after the contract period, customer behavior becomes less predictable with the same set of variables, leading to reduced predictive accuracy.

12-month product type

The 12-month product type models exhibited a performance pattern like the 6-month product models, with a higher concordance index score observed before the 12-month period and a lower score after the 12-month period (see Table 1). To construct the 12-month product type dataset, the full dataset was segmented using the product type variable, which identified the associated product type for each observation. To enable a more detailed analysis of customer behavior, the 12-month dataset was further divided into two subgroups: "before 12-month" and "after 12-month" periods.

Segmenting the full dataset by product type revealed significant performance improvements for the 6-month product

type before the contract ends, while the 12-month product type exhibited a similar pattern but with relatively more stable performance. This analysis highlights two key insights in the context of the CPH model.

- **Customer-Based Model:** A model designed to predict individual customer behavior should account for the substantial influence of product type on churn rates. Incorporating product-specific factors allows for actionable insights tailored to individual customer behaviors and decisions.
- **Product-Based Model**: A model aimed at predicting trends in churn rates across different products benefits significantly from segmenting the dataset by product type. Customers in various product groups exhibit distinct churn rate patterns, as shown by the CPH models above. These variations can have a profound impact on model performance and should be carefully considered during analysis to ensure accurate and meaningful predictions.

This approach highlights the importance of tailoring analytical strategies to the unique characteristics of customer and product segments

Aalen additive model

Before fitting the AA model, quantitative features were discretized and dummy coded as described in the Materials and Methods sections. These features included Heating Degree Days, Average Daily Utility Usage, Product Switch Count, Utility Price at Enrollment, Aggressive 6-month Rate at Enrollment, Aggressive 12-month Rate at Enrollment, Standard 6-month Rate at Enrollment, Standard 12-month Rate at Enrollment, NYMEX at Enrollment.

Like the approach used for the CPH model, the dataset was segmented into subsets of customers with 6-month or 12-month contracts, then further divided into "before" and "after" the end of their product contracts. This segmentation allowed the model to effectively capture and analyze time-dependent customer behavior.

During training of the Aalen Additive (AA) model, a grid search over 10 values of the regularization parameter τ such that the hyperparameter τ is in [0.0001, 0.0004, 0.002, 0.01, 0.05, 0.20, 1, 5, 20, 100], evenly spaced on a log scale. Combined with fivefold cross-validation, we determined the optimal hyperparameter value for each customer subgroup. The hyperparameters that produced the highest concordance index score in each customer subset was selected for the model.

Table 2 presents the training concordance index scores of the AA model for each data set subset, along with the corresponding optimal hyperparameters based on the average concordance index scores using fivefold cross validation.

Tabel 2 Training concordance index scores for the Aalen Additive model

Customer Groups	Optimal Hyperparameter (τ)	AA Training Concordance Index Score
6-month contract customers before the 6th month	20	76.62%
6-month contract customers after the 6th month	100	74.68%
12-month contract customers before the 12th month	100	72.68%
12-month contract customers after the 12th month	20	79.30%

Tabel 3 Testing concordance index scores for the Aalen Additive model

Customer Groups	Optimal Hyperparameter (τ)	AA Testing Concordance Index Score
6-month contract customers before the 6th month	20	72.20%
6-month contract customers after the 6th month	100	73.99%
12-month contract customers before the 12th month	100	70.19%
12-month contract customers after the 12th month	100	80.70%

The testing concordance index scores for the AA model are presented in Table 3. The results are consistent with the training outcomes: the highest test concordance index is observed among 12-month contract customers evaluated after their contract period. In contrast, the subset of 12-month customers assessed *before* contract expiration achieved the lowest concordance index among the four groups, though still demonstrating strong predictive performance with a score of 70.19%. The 6-month contract customers also show comparable performance between the training and testing datasets, confirming the model's robustness in analyzing customer churn both before and after contract expiration, as it effectively captures behavioral patterns throughout the active contract period.

To evaluate how the risk of customer churn accumulates over time across various customer segments, we plotted cumulative hazard functions for several key predictors. Specifically, the effects of Heating Degree Days, Average Daily Utility Usage, and Product Switch Count were visualized for selected customer subsets. These variables were chosen to illustrate the interpretability of the AA model while maintaining clarity in presentation, plotting all predictors would have produced overly complex figures.



Specifically, cumulative hazard functions for customers with 6th-month contracts (before and after their contract period) are provided in Figs. 6a and 7a (Appendix). Corresponding results for customers with 12th-month contracts (also before and after their contract period) are provided in Figs. 8a and 9a (Appendix). In addition, survival functions for several randomly selected individual customers were plotted to illustrate how churn risk develops at the customer level by prediction. These individual survival functions, for customers chosen within each customer subgroup, are displayed in Figs. 6b, 7b, 8b and 9b in the Appendix section. The following paragraphs discuss these results in greater detail, focusing on how churn risk changes before and after contract periods and which predictors most strongly influenced customer behavior.

In the context of the AA model, we use "significant drivers" to refer to factors whose effects are evident from the cumulative hazard coefficient function plots (Figs. 6a, 7a, 8a, and 9a). These functions represent the cumulative effect of a predictor on the hazard of churn over time. A trend diverging from zero indicates a meaningful time-varying effect, while a stable, flat curve close to zero suggests little or no effect.

For example, in Fig. 6a (6-month contract customers prior to expiration), we observed that all predictors show a steadily increasing trend from the 4th month to the 5th month, indicating the key time all these factors are driving the churn. In contrast, in the beginning months, all curves remained relatively flat and close to zero, meaning no significant effect during this period.

For 12-month contracts, Fig. 8a in the Appendix section illustrates factor effects before contact expiration. As shown, the cumulative hazard curves for the predictors remain flat up to approximately the 5th month, indicating minimal influence on customer churn during this period. Throughout most of the contract duration, the cumulative hazard trajectories stay relatively flat, with only a slight increase observed toward the contract's end.

However, for 12-month contract customers continuing beyond 12 months, Fig. 9a in the Appendix section shows *Median Heating Degree Days* and *Low Standard Month Rate* as a significant churn driver. This is evident from the steady rise in the cumulative hazard curve for this predictor, peaking around the 18th month. Conversely, predictors with falling cumulative hazard curves, such as *Heating Degree Days (high category)*, suggest that their contribution to the hazard rate is markedly less pronounced. This could suggest that customers in regions with extreme cold temperatures are less likely to switch providers, possibly due to high reliance on service, fewer alternatives, or the perceived risk of switching. These observations highlight the dynamic impact of specific predictors on customer churn at different stages of the contract period.

Finally, customer-level predictions across all groups demonstrate that the AA model is effective in identifying individuals at high risk of churn, as shown in Figs. 6b, 7b, 8b, and 9b in the Appendix. For example, Figs. 7b and 9b clearly show that customers can be segmented into two distinct groups: those with high churn risk and those with low churn risk in the beginning 12 to 18 months. These insights highlight the model's practical value for customer retention strategies, enabling businesses to pinpoint vulnerable customers and intervene proactively to reduce attrition.

The AA model's ability to provide survival predictions for individual customers is a notable advantage over the CPH model. This feature allows for more personalized customer insights, enabling targeted retention strategies based on each customer's unique churn risk profile.

Discussion

Additional insights from CPH and AA models

For both the CPH and AA models, *Product Type* emerged as a critical variable. Customers displayed distinct behaviors at predefined contract intervals (6 months and 12 months), which significantly impacted the overall performance of both models. These contract milestones shaped customer retention patterns, with notable differences in churn risk depending on the contract duration and the period relative to contract expiration.

However, when comparing model performance before and after the contract period, it appears that the CPH model better predicts customer churn before the contract expires, while the AA model performs well across all product contract periods. However, this performance pattern may be specific to the scenario presented by the dataset being used.

For the CPH models, the significant performance drop after the contract expiration may be attributed to data availability: the "before" models include comprehensive information about customers prior to the end of their contracts, while the "after" models lose substantial data on customers who have already churned. This loss of information likely makes it more challenging for the "after" models to accurately forecast survival rates compared to the "before" models.

For the AA models, time-varying information plays a more significant role during the post-contract period, making the AA model particularly well-suited for analyzing customer behavior after contracts expire. Additionally, the hyperparameter settings are seen to differ across customer segments, depending on the dataset type (before and after the end of contract period). This variation arises due to differences in data structure, the intricacy of the model, and



the randomness inherent in training machine learning models. Indeed, randomness in weight initialization or mini-batch sampling has been seen to lead to different optimal configurations across runs in machine learning models (Bengio 2012).

This paper focuses on applying survival analysis models to customer churn prediction, with the primary goal of demonstrating their interpretability and effectiveness in capturing time-to-event dynamics. A more systematic exploration of hyperparameter optimization could be pursued as a future direction, building on existing work such as (Wang and Ni 2019), which highlights the role of Bayesian hyperparameter optimization in improving predictive model performance.

It is important to note that the performance patterns observed in this study may not generalize across all datasets or business contexts. Model results are highly dependent on the specific characteristics of the data, such as contract structure, customer mix, or market conditions. A key takeaway, however, is that building product-level churn models can be more effective than relying solely on one model for all customers, as demonstrated by the significant performance differences between the two in this study. This finding underscores the value of tailoring churn prediction models to the granularity of individual products or services, which can provide more precise insights and actionable strategies for customer retention.

In summary, both the CPH model and the AA model have proven to be reliable tools for predicting customer churn. Each model provides value by forecasting survival rates over time, offering users critical insights to develop proactive strategies that mitigate churn and enhance customer retention efforts.

Practical retention strategies

The analysis presented in this paper highlights opportunities for companies to develop targeted, time-sensitive customer retention strategies. By leveraging insights from survival analysis, utility firms – and other industries facing similar conditions – can design interventions tailored to customer risk profiles and contract timelines. Here are a few suggestions:

Customer segmentation guided by survival risk analysis

Customers can be segmented based on their probability of churn, allowing decision makers to prioritize interventions. Survival and hazard rate patterns can suggest distinct risk categories, such as:

Critical Risk: likely to churn within 30 days
High Risk: likely to churn within 30–60 days
Moderate Risk: likely to churn within 60–90 days
Low Risk: likely to churn after 90 + days

This segmentation framework provides managers with actionable intelligence on which groups require immediate retention efforts versus long-term engagement strategies.

Contract time critical intervention program

The survival curves illustrate sharp declines in customer survival probability at the 6-month and 12-month contract points, underscoring these as high-risk renewal periods. Utility firms could implement campaigns that attract customers at earlier stages of the contract lifetime (e.g. 60 days before contract ends). Some of the campaigns could involve:

- Offering personalized retention package based on individual customer hazard rates.
- Highlighting usage-based value propositions for high-consumption customers.
- Introducing budget-friendly alternatives or optimization tools for low-consumption customers.

These time-sensitive strategies demonstrate how survival analysis goes beyond static churn probabilities by revealing not only which factors drive churn but also *when* interventions should occur, thereby outperforming traditional models that lack this temporal and explanatory precision.

Conclusion

This study highlights the effectiveness of both the Cox Proportional Hazards (CPH) and Aalen Additive (AA) models in predicting customer churn based on product type. The CPH model achieved concordance index scores of 72% for the 6-month product and 79% for the 12-month product, demonstrating strong performance in identifying churn risk in these groups. Similarly, the AA model showcased its utility with concordance index scores of 60.96% for the 6-month product type before contract expiration and 74.45% after expiration. For the 12-month product, the AA model achieved concordance index scores of 58.85% before contract expiration and 78.65% after expiration, emphasizing its ability to capture churn dynamics more effectively in the post-contract period. These results highlight the strengths of each model at different phases of the contract lifecycle, while recognizing that performance may vary depending on the dataset, as discussed earlier.



Beyond predictive accuracy, the cumulative hazard trajectories provided by the AA model identified key covariates driving churn risk, offering valuable interpretability and precision at the individual customer level. This ability to link predictors to time-varying churn risk demonstrates how survival analysis moves beyond static probability models by capturing when churn is most likely to occur.

Analysis of churn patterns further revealed that product renewal dates are especially high-risk periods, making them critical targets for retention efforts. Strategically timed interventions – such as renewal discounts, seasonal pricing adjustments, and loyalty programs (Nitin et al. 2023) – can be designed directly from these survival insights. In this way, observed hazard patterns translate into actionable CRM (customer retention management) strategies that address customer attrition at its most vulnerable points.

Overall, the findings confirm that survival analysis models provide businesses with a time-sensitive, data-driven framework for churn prevention. By enabling segmentation, prioritization, and precise timing of interventions, survival models outperform traditional churn prediction approaches. These strengths make them invaluable not only for customer retention strategies but also for broader applications such as uplift modeling, propensity scoring, and customer lifetime value analysis.

Appendix

See Figs. 6, 7, 8, and 9; Table 4.

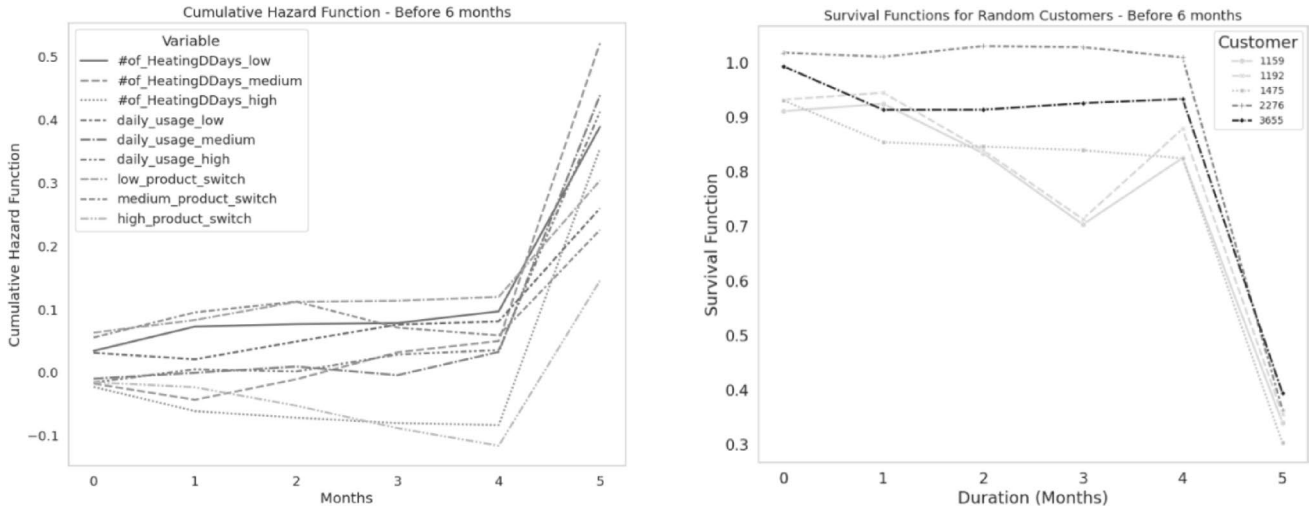


Fig. 6 **a** and **b** Cumulative hazard coefficient functions for 6-month contract customers before the 6th month of the contract and customer-level predictions for several randomly selected 6-month contract customers during the same period

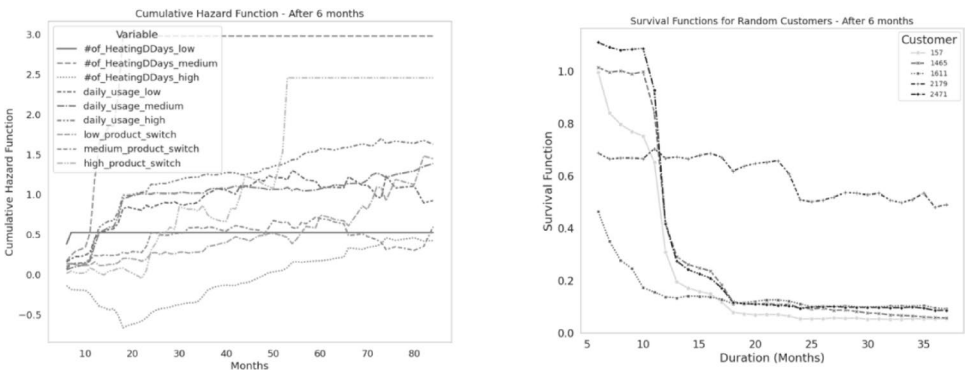


Fig. 7 **a** and **b** Cumulative hazard coefficient functions for 6-month contract customers after the 6th month of the contract and customer-level predictions for several randomly selected 6-month contract customers during the same period

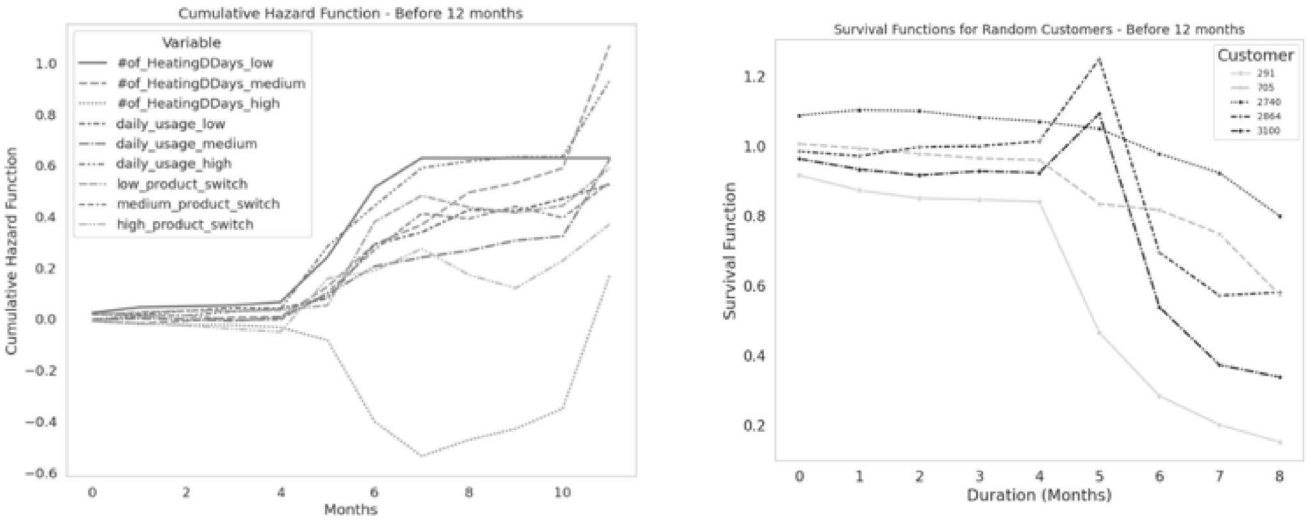


Fig. 8 **a** and **b** Cumulative hazard coefficient functions for 12-month contract customers before the 12th month of the contract, and customer-level predictions for several randomly selected 12-month contract customers during the same period

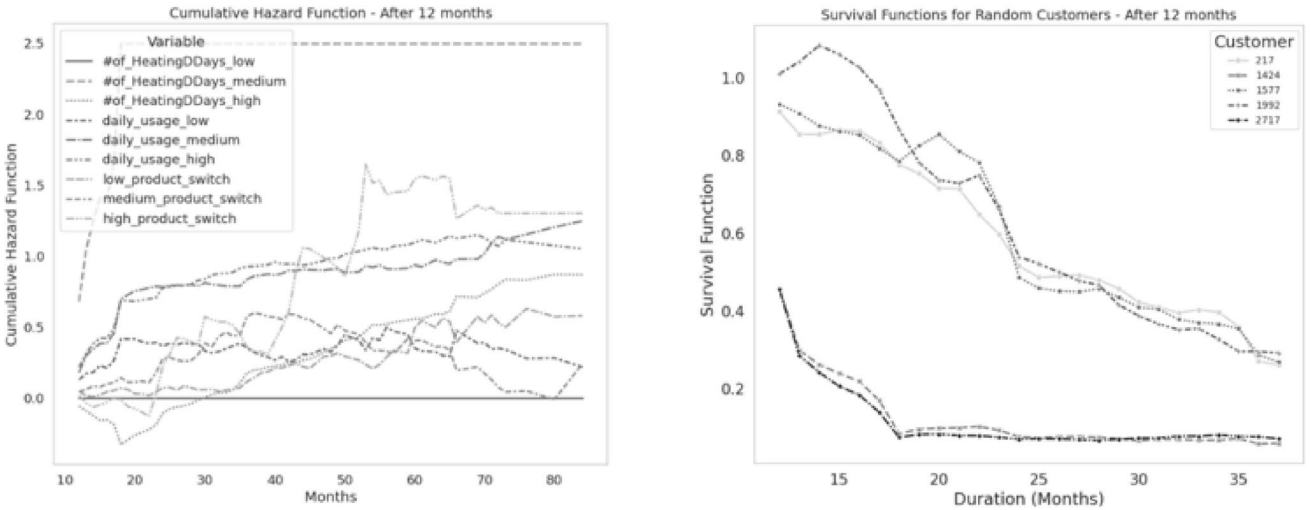


Fig. 9 **a** and **b** Cumulative hazard coefficient functions for 12-month contract customers after the 12th month of the contract, and customer-level predictions for several randomly selected 12-month contract customers during the same period

Table 4 Final variables selected for survival analysis

Variables selected for the survival analysis experiments		Description of variable
1	ACR Flag	The automatic contract renewal (ACR) flag: 1 for ON and 0 for OFF
2	Utility Price at Enrollment	The utility price at the time of customer enrollment
3	Aggressive 6 M Rate at Enrollment	The low promotional rate offered to new six-month customers to encourage enrollment
4	Aggressive 12 M Rate at Enrollment	The low promotional rate offered to new twelve-month customers to encourage enrollment
5	Median Household Income	The median income for households in the customer's address area, defined by the 9-digit ZIP (ZIP+4) geography
6	NYMEX at Enrollment ($/CCF)	The utility rate tied to the New York Mercantile Exchange (NYMEX) pricing at the time of enrollment
7	Standard 6 M Rate at Enrollment	The utility rate guaranteed by the utility provider for 6-month contracts at the time of enrollment
8	Standard 12 M Rate at Enrollment	The utility rate guaranteed by the utility provider for 12-month contracts at the time of enrollment
9	HDD65	The number of heating degree days in a month with average temperature below 65°F
10	Daily Usage	The customer's average daily utility usage amount in a month
11	Product Plan Switch	The number of product plan switches (number of times a customer has switched plans internally)
12	Duration (Months)	The length of time (in months) from a defined starting point until customer churn occurs or the customer is censored
13	Churn Flag	The binary variable (1 or 0), indicates whether a customer churned (=1) or not (=0)

The AA model uses discretized features, as outlined in the manuscript's Materials and Methods section. For example, HDD65 is discretized into four levels, i.e. very low, low, medium and high. In the AA model the very low category is used as the reference level and the other three categories are coded with binary dummy variables

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